CHAPTER

14

Mental Models and Conceptual Design

Highlights

- Mental models
  - What is a mental model?
  - The ideal mental model
  - The designer's mental model
  - The user's mental model
  - The conceptual design as a mapping between mental models
- Making the conceptual design
  - Start with a conceptual design for the ecology
  - Conceptual design for interaction
  - Conceptual design for emotional impact
  - Leverage design patterns in conceptual design
  - Leverage metaphors in conceptual design
  - Conceptual design for sub-systems by work role

14.1 INTRODUCTION

14.1.1 You Are Here

We begin each process chapter with a "You are here" picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 14.1). In this chapter, we continue the UX design process with mental models that designers and users have about how a system works and conceptual designs that connect their mental models.

14.1.2 Mental Models

A mental model in UX is a description, understanding, or explanation of someone's (e.g., designer or user) thought process about how a product or system works. If a user's mental model of how a product or system works is

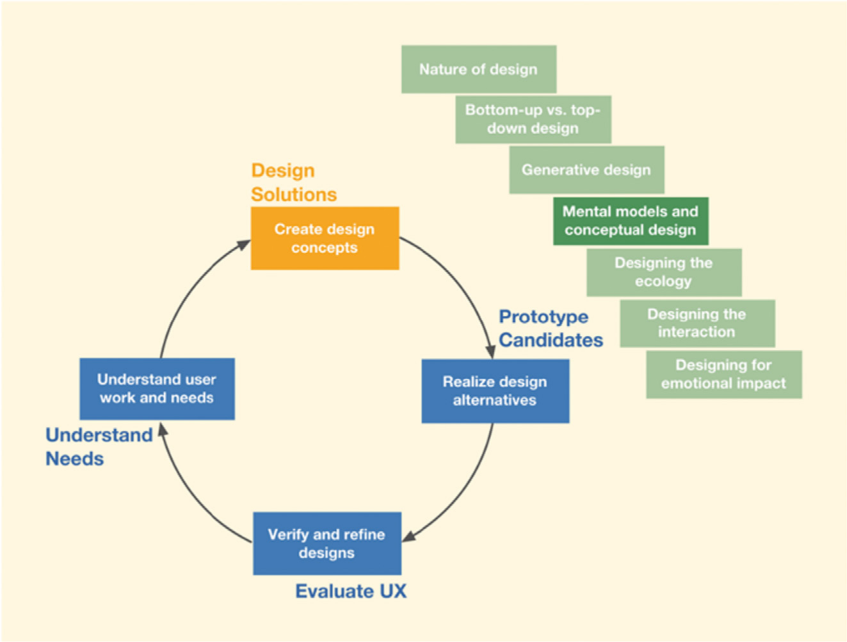

Fig. 14.1

You are here in the chapter describing the role of mental models and conceptual design within the design solutions activity in the context of the overall Wheel UX lifecycle.

correct, the user will know how to use the system. It is up to the designer to create a conceptual design capable of conveying a correct mental model to users.

14.2 HOW A CONCEPTUAL DESIGN WORKS AS A CONNECTION OF MENTAL MODELS

Fig. 14.2 shows the conceptual design (in the center of the figure) as a kind of mapping from the designer's mental model (near the top) to the user's mental model (near the bottom).

Here's how it works:

1. The ideal mental model is an abstraction that represents the reality of how something like a thermostat works out in the world.
2. The designer studies this reality via UX research, interactions with subject-matter experts, analysis, etc.
3. The designer develops a (possibly partially correct) mental model.
4. The designer builds this mental model into the conceptual design. This step is your challenge!
5. The conceptual design conveys that knowledge of the designer's understanding to users.

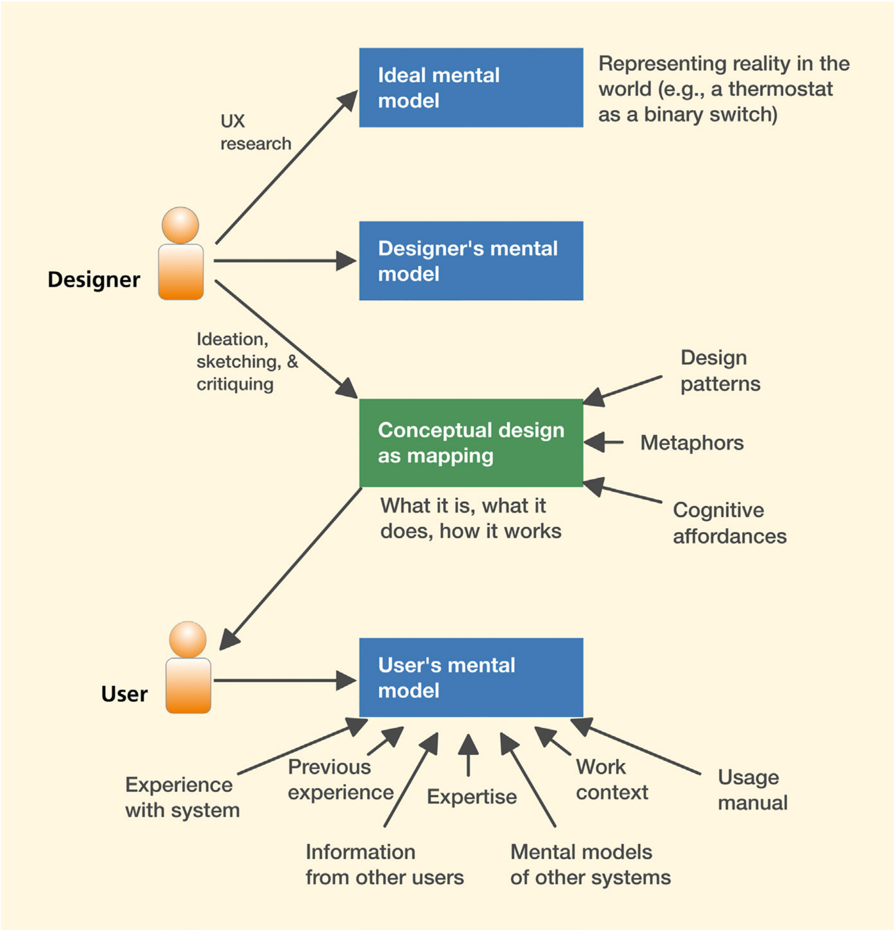

Fig. 14.2
The conceptual design as a mapping of the designer's mental model to the user's mental model.

6. If the user has some a priori understanding of the reality, the conceptual design can either affirm or challenge that understanding. Otherwise, the user (hopefully) learns how this system or product works from the conceptual design.

Before we try to explain this whole concept further, we ask that you, the reader, indulge us by reading through Section 14.2.3, on the user's mental model, to get a brief description of each of the four boxes in Fig. 14.2. That should give you the vocabulary for an intuitive explanation.

14.2.1 The Ideal Mental Model

An ideal mental model is a hypothetical correct description of the reality of a system or product and how it works. Continuing with a common house thermostat as an example, the user's manual would probably say something to the effect that you turn it up or down to make it warmer or cooler, but it would probably fall short of the full explanation of how a thermostat works.

Thermostats are very simple devices, and their ideal mental model is widely known and understood. Most thermostats, as Norman (1990) explains, are binary switches that are simply either on or off. During winter, when the sensed ambient temperature is below the target value, the switch closes, and the thermostat turns on the heat. When the temperature then climbs to the target value, the heating system switches open, and the thermostat turns off the heat source. It is therefore an incorrect mental model to believe (taken just as an example) that you can make a room warm up faster by turning the thermostat up higher than the target temperature when, in fact, that action would only make the final target temperature higher. Such an incorrect mental model could arise from other correct mental models that are similar but depend on a slightly different device operation. For example, the accelerator in a car is part of what is used to control the speed. If you want to reach a target speed, you depress the accelerator to a certain point; however, if you want to reach the target speed faster, you would temporarily press the accelerator well beyond the target point, and the engine would race to reach that speed much faster. (Thanks to Prof. Panagiotis Apostolellis for this example.)

14.2.2 The Designer's Mental Model

Simply put, the designer's mental model (the second box in Fig. 14.2) is the UX designer's conceptualization of the ideal mental model. The designer studies the ideal mental model, the UX research results, and data models and requirements to visualize what to incorporate into the conceptual design.

14.2.3 The User's Mental Model

Veer and Melguizo (2003), citing Carroll and Olson (1987), define a user's mental model (the fourth box in Fig. 14.2) as the "mental representation that reflects the user's understanding of the system." As Norman (1990) says, it is a natural human response to an unfamiliar situation to begin building an explanatory model a piece at a time. We look for cause-and-effect relationships and form theories to explain what we observe and why, which then helps guide further behavior and actions in task performance.

As shown in Fig. 14.2, each user's mental model is a product of many different inputs, including, as Norman has often said, knowledge in the head and knowledge in the world. Knowledge in the head comes from mental models of other systems, user expertise, and previous experience. Knowledge in the world comes from other users, work context, shared cultural conventions, documentation, and the conceptual design of the system itself. This latter source of user knowledge is the responsibility of the UX designer.

14.2.4 The Conceptual Design as Mapping Between Mental Models

A conceptual design (third box in Fig. 14.2) is the part of a design containing a theme, metaphor, notion, or idea with the purpose of communicating a design vision about a system or product, corresponding to what Norman (1990) called the "system image" of the designer's mental model (p. 16). The goal of a conceptual design is to communicate the designer's mental model to users so they will know how to use the system. Without an effective conceptual design, users cannot leverage any experience they gain from interacting with one part of a system or with other systems. It is not uncommon, however, for systems to be built without a clear conceptual design upfront. The result can be a poorly focused design that fails as a mapping, thus frustrating users.

Example: The Washington Metro Ticket Kiosk

The Washington, DC, metro ticket kiosk (Fig. 14.3), circa 2018, which does not look like any other kiosk, is an example of a system that lacks a good conceptual design. It is what we call a "show up and throw up" user interface. This type of design is also sometimes called information flooding. The designers failed to determine what information users needed and when, so they decided to show everything at once and let the users figure it out from there.

A bewildering array of buttons and displays meets the user who walks up to this kiosk in a hurry to purchase a fare to get on the train. Additionally, a significant proportion of these users are tourists, who have no prior experience with this system or even, perhaps, a good grasp of the English language. Also, there is usually a line of people waiting behind the current user, so the pressure is on to complete the transaction and not hold up the line.

When users attempt to purchase a one-time fare card, they first have to figure out the cost of the fare by reading the miniscule text at the top of the kiosk. If they are successful, then they have to select one of the three options next to the "Select Purchase" section at the top to get started with the type of card they are purchasing. This selection, thankfully, is near the top of the screen, a reasonable location the user will look for to get started. However, if users already have a rechargeable Smartrip fare card, then they have to start by tapping their card on the circular widget in the bottom center part of the kiosk, to the lower right of the "Insert Payment" label. Then they have to select an option from the "Select Purchase" step at the top. Then they pay using one of the available options in the center, and touch the circular object in the

Fig. 14.3
A Washington, DC, metro ticket kiosk.

center again. It is an arbitrary sequence of steps that is only explained through small text in the center of the kiosk.

Even if designers had a clear mental model, they failed to articulate it in an effective conceptual design; consequently, the users have no way of forming a mental model of how the system works. The three steps of the workflow are not supported consistently for all types of transactions. For example, Smartrip users or users trying to combine their existing fare card balances into a single card do not start in the "Select Purchase" step. Similarly, not all payment steps are organized in the payment section in the middle (note the coin return at the bottom). Because people often struggled with this process, the Metro system had to post (and pay for) metro employees onsite to help, which defeats the purpose of having a kiosk in the first place, right?

Compare this to the New York City ticket kiosk, where the user is guided through a task flow user interface and is not overwhelmed with all options at once. This kiosk (Fig. 14.4) uses a touchscreen and has a single option for

Fig. 14.4

New York City subway kiosk with a touchscreen. (From Dr. Shahtab Wahid, UX Team Lead, Terminal & Web Platforms, Bloomberg, LP.)

users to initiate their transaction: Select the "Start" button next to a clear cognitive affordance "Touch start to begin." Fig. 14.5 shows a closeup of the touchscreen part of the kiosk.

The user is then prompted for language preference (Fig. 14.6), followed by the type of fare card to purchase (Fig. 14.7) and payment method (Fig. 14.8).

The broader conceptual design of this system (i.e., a step-by-step dialog involving one task at a time) is better articulated and communicated through a design that shows information relevant to the user's sequence of tasks when it is needed and not all at once, as in the Washington, DC, metro system.

Fig. 14.5

"Touch start to begin" as starting point for the ticket buying transaction. (From Dr. Shahtab Wahid, UX Team Lead, Terminal & Web Platforms, Bloomberg, LP.)

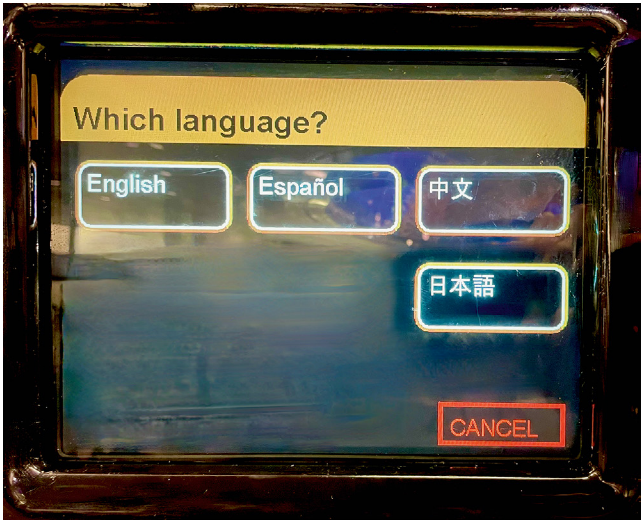

Fig. 14.6

Prompt to select language. (From Dr. Shahtab Wahid, UX Team Lead, Terminal & Web Platforms, Bloomberg, LP.)

Fig. 14.7

Prompt to select type of fare card. (From Dr. Shahtab Wahid, UX Team Lead, Terminal & Web Platforms, Bloomberg, LP.)

Fig. 14.8

Prompt to select payment method. (From Dr. Shahtab Wahid, UX Team Lead, Terminal & Web Platforms, Bloomberg, LP.)

14.2.5 The Nature of the Mental Models and Conceptual Design

What do the four boxes in Fig. 14.2 indicate?

- The top box, the ideal mental model, is a hypothetical abstraction of an accurate version of how something works.
- The second box, the designer's mental model, would be the same as the top box if the designer's understanding of how the thing works is correct.
- The first, second, and fourth (the user's mental model) boxes represent mental models. They are internalizations (i.e., abstract schemas that exist inside the brain). It is the stuff of sense making and how we encode, organize, and store information in

long-term memory. The UX designer has to know and understand these models, but the focus is on the third box.

- The third box (conceptual design as mapping) differs from the other three because it involves design, not a model. A conceptual design is an externalization of how something works, as encoded in a design. Creating the conceptual design is the job of the UX designer.

14.2.6 Creating the Conceptual Design: A Design Challenge

The conceptual design is built by incorporating the designer's mental model, as an explanation of how the system works, into the design. This topic can be difficult for students and new UX practitioners. This is the only box that is tangible, and it is the one users see in the product or system. There are practically no limits to the ways it can be expressed. There is no standard format; it is customizable to each situation and according to your conceptual design. Following are ways to build an explanation of product development:

- Well-worded textual description of how it works
- Schematic diagram
- Concept map
- Type of model
- Flow chart
- Animation that shows moving parts
- Compelling analogy
- Appeal to a mutually understood metaphor
- Example of a generic user's journey
- Stepwise wizard guiding users through a task sequence
- Instructional video via YouTube

For example, if a thermostat displays a schematic of the cooling setup and shows via some animations how the sensor checks the ambient temperature and compares it with the set temperature, and then proceeds to open or close the switch, that would be the conceptual design, an expression of the designer's mental model in the design itself.

Conceptual design is where you innovate and brainstorm to plant and first nurture the user experience seed to build this understanding. Without an effective conceptual design upfront, it is much harder to iterate the design later to yield a good user experience. Conceptual design is where you establish the metaphor or the theme of the product—in a word, the concept. To paraphrase the Johnson and Henderson (2002) rule for conceptual design, if something is not in the conceptual design, then the system should not require users to be aware of it.

Example: How Mental Models Can Make for Entertainment

Lack of a correct user mental model can be the stuff of comedy curve balls. An example is the scene in the 1992 movie, *My Cousin Vinny*, where Marisa Tomei—as Vinny's fiancée, Mona Lisa Vito—tries to make a simple phone call. This fish-out-of-water scene pits a brash young woman from New York against a rotary dial telephone in a backwater southern town. You cannot help but reflect on the mismatch in the mapping between her mental model of touchtone operation and the reality of old-fashioned rotary dials as she pokes vigorously at the numbers through the finger holes.

Lest you dismiss her as completely out of touch, however, we remind you that it was she who solved the case with her esoteric automobile knowledge, proving that Vinny's mint green 1964 Buick Skylark convertible was not the getaway car. The concept of a limited-slip differential is the crucial element of the mental model of how a car works on which the whole story hinges. In the ideal mental model, a limited-slip differential couples the two rear wheels together under acceleration. Because Vinnie's car did not have this feature, it could not have been the car that made the two parallel tire marks left at the scene of the crime. Hence the verdict: not guilty!

14.3 UX DESIGN STARTS WITH CONCEPTUAL DESIGN

After doing UX research, many designers jump right into sketching out screens, menu structures, and widgets. Johnson and Henderson (2002), however, will tell you to begin with conceptual design before sketching any screen or user interface objects. As they put it, screen sketches are designs of "how the system presents itself to users. It is better to start by designing what the system is to them." A clear and consistent conceptual design ensures that the rest of the product or system design presents a unified and coherent appearance to the user.

Screen designs and widgets will come, but time and effort spent on interaction details can be wasted without a well-defined underlying conceptual structure. Norman (2008) puts it this way: "What people want is usable devices, *which translates into understandable ones* [emphasis ours]."

14.3.1 Need for a Conceptual Design Component at Every Level in the User Needs Pyramid

Before we begin to design for the ecology, interaction, and emotional impact, we must create a component of the conceptual design for each layer in the user needs pyramid (see Section 11.2.1) (Fig. 14.9):

- An ecological component that helps users understand how the product or system fits into its broader ecology and works together with other products and systems in that ecology
- An interaction component that helps users understand how to use the product or system
- An emotional component that conveys the intended emotional impact

The components of the conceptual design that address these three layers are described in the following sections.

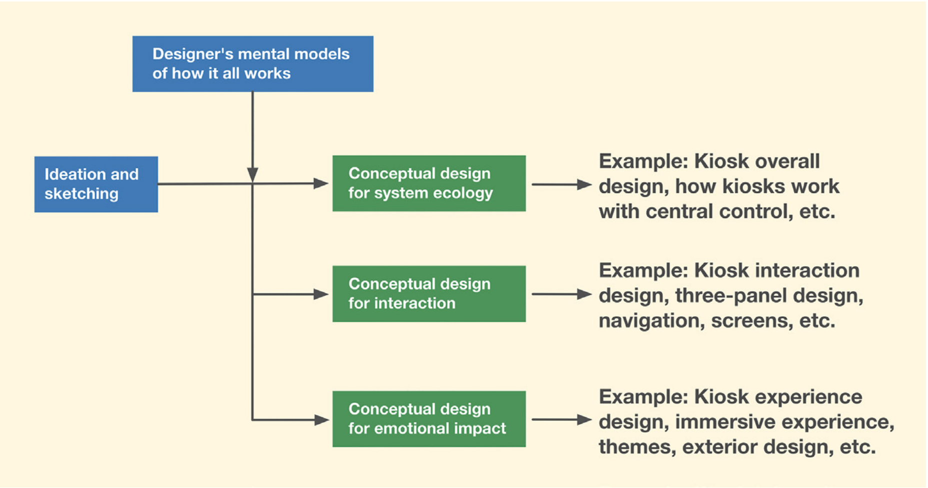

Fig. 14.9

Designer workflow and connections among the three layers of the user needs pyramid.

14.3.2 Conceptual Design for Work Practice Ecology: Describing Full Usage Context

The mental model for the system ecology helps users understand how the product or system fits into its ecology and, as the user interacts with products and systems in that ecology, how the system fits into the life of the user. Let us expand the explanation of home thermostats to include their ecological setting. For example, is the system to be used at home, on a desktop/laptop/smartphone, on the go, while traveling/working/shopping?

A thermostat ecology includes a heating (and/or cooling) system consisting of three major parts: a heating source or cooling unit, a heating or cooling distribution network, and a control unit, the latter being the thermostat and some other hidden circuitry. The heat source could be, for example, gas or electric. The cooling unit may use electric power to run a compressor. The distribution network would use fans to send heated or cooled air through air ducts, or a pump would send heated or cooled water through sub-floor pipes. There is also a living space being heated or cooled and its ambient temperature, plus a human user.

Next, we address what a thermostat does within its ecology. It mediates the temperature in a room or other space by turning on the heating or cooling source until the temperature in the living space stays near a user-settable value to keep people comfortable.

If the designers are planning to create a smart thermostat, then their mental model may also include a way to sense users in the house to ascertain occupancy, conserving energy when the house is unoccupied. This includes sensor locations, how data from those sensors will be communicated to the thermostat, and how users will be able to control the system when they are away from the house. Thus, the conceptual design may expand to include internet connectivity, whether the data are saved over a period of time for trend analysis, and whether the system provides other types of feedback, such as comparing with energy consumption averages in the neighborhood or other grouping.

14.3.3 Conceptual Design for Interaction: Describing How Users Will Operate It

For interaction, a conceptual design is a task-oriented view about how a user operates the system or product. Different platforms or interaction devices call for different interaction models and corresponding conceptual designs. For example, the thermostat could be a traditional wall-mounted physical device or a smartphone app that interfaces with the physical device via Wi-Fi or the internet.

In the physical device case, perhaps a user can see two numeric temperature displays, either analog or digital. One value is for the current ambient temperature, and the other is the setting for the target temperature. There will be a rotatable knob, button, slider, or other interaction device to set the desired target temperature. Other interactive features could include programming a weekly schedule, which goes well beyond turning a knob or pressing buttons to change the temperature.

Design of the interaction should focus on remedying anticipated breakdowns in the user's mental model, where there is uncertainty about what actions to take to perform routine tasks. Close attention to affordances (see Chapter 27), especially cognitive affordances, in the overall UX design will aid this mapping and help users make use of the conceptual design while avoiding breakdowns.

14.3.4 Conceptual Design in the Emotional Perspective: Describing Intended Emotional Impact

For emotional aspects, the conceptual design is a description of the expected overarching emotional response. Regarding the thermostat example, this could be about the emotional effect of a modern and sleek aesthetic with steel and glass components in contrast to traditional designs made with plastic materials. The conceptual design for emotion will also include how the physical design fits in with the house décor and the craftsmanship of its construction. The designers may also have specific plans for the visual design of the display on the device, including the type of light-emitting diode to use, backlighting options, and whether there will be color options available. A traditional plain appearance gives up emotional ground to the sophisticated look of a Nest thermostat (Fig. 14.10), especially when that stylish exterior carries with it the knowledge that the Nest is really cool because of what it can do inside. Maybe a thermostat design could be made to emote a feeling of coziness with being warm or cool enough to be very comfortable.

A B

Fig. 14.10

The appearance of an old thermostat versus that of a versus that of a Nest (Image source: Google).

14.3.5 Leveraging Design Patterns in Conceptual Design

A design pattern is a repeatable solution to a common design problem that emerges as a best practice, encouraging sharing, reuse, and consistency (see Sections 16.3.4 and 16.8). Design patterns are design styles that entail reuse of interaction structures, color schemes, fonts, physical layout, look and feel, and locations of interaction objects. They help with consistency and ease of learning. As an example of designers leveraging the design patterns of conceptual designs from known applications to new ones, consider a well-known application, such as Microsoft Outlook. People are familiar with the navigation bar on the left side, the list view at the top right side, and a preview of the selected item below the list. When designers use that same idea in the conceptual design of a new mail application, the familiarity carries over.

14.3.6 Leveraging Metaphors to Carry Meaning in Conceptual Design

A metaphor is an analogy used in design to communicate and explain unfamiliar concepts using familiar conventional knowledge. A central metaphor often becomes the theme of a product, the concept behind the conceptual design. What users already know about an existing system or existing phenomena can help them learn how to use a new system (Carroll, Mack, & Kellogg, 1988; Carroll & Thomas, 1982).

A simple example is the metaphor of reading a book on an iPad. As the user moves a finger across the display to push the page aside, the display takes on the appearance of a real page turning. Most users find it comfortingly familiar.

A more extensive example is the now-pervasive desktop metaphor. When the idea of graphical user interfaces in personal computers became an economic feasibility, the designers at Xerox Parc were faced with an interesting UX design challenge: how to communicate to the users, most of whom were going to see this kind of computer for the first time, the way interaction design works.

In response, they created the powerful desktop metaphor. The design leveraged the familiarity people had with how a desktop works: It has files, folders, a space where current work documents are placed, and a recycle bin where documents can be discarded (and later recovered, until the bin itself is emptied). This analogy of a simple everyday desk was brilliant in its simplicity and made it possible to communicate the complexity of a new technology.

Another great example of a metaphor for interaction conceptual design can be found in the Time Machine feature on the Macintosh operating system. It is a backup feature where the user can take a time machine and, via animation, fly back in time to visit older backups for retrieving lost or accidentally deleted files.

Any files from a backup can be selected and brought back to the present to recover lost files. This conceptual design is effective because it uses a familiar metaphor to simplify and explain the complexity of a somewhat technical and cumbersome activity of data backup and recovery to everyday users. Other backup products provide similar capabilities but are more difficult to interact with because they lack a similar metaphor to reflect how it works.

Example: Physics as a Metaphor in UX Conceptual Designs

Scrolling through lists on an iOS device, such as an iPad, done by swiping the finger up or down, demonstrates a phenomenon called physics in interaction. The scrolling is done as though the display were on a large wheel that you spin with your finger. This display appears to exhibit significant mass, so it has a kind of virtual inertia that keeps it spinning after you release your finger, but it also exhibits friction that soon slows the display to a stop.

This display involves mass, inertia, and friction, which are parameters of physics. The way the mass and inertia are exhibited is so real that it actually feels like a wheel to the finger. It is a kind of artificial physicality. If users spin it past where they want to be, then the move is technically considered an error by most definitions, yet it does not feel like an error. It feels normal for a wheel with mass, and users naturally spin it back a bit.

14.3.6.1 Metaphors can cause confusion if not used properly

Metaphors, like any analogy, can break down when the existing knowledge and the new design do not match. When a metaphor breaks down, it is a violation of the agreement implicit in a conceptual design. The famous criticism of the Macintosh platform's design of ejecting an external disk by dragging its icon into the Trash is a well-known illustration of how a metaphor breakdown attracts attention. It may have been more faithful to the desktop metaphor if the system would delete the contents of an external disk when it is dragged and dropped into the Trash instead of ejecting it; however, that would be too dangerous. Perhaps ejection could have been better portrayed by popping a device symbol out of a computer symbol.

We cover designing (including creating conceptual designs) for each of the layers of the pyramid in the next three chapters.

Exercise 14.1 Conceptual Design for Your Product or System

Goal: Practice initial conceptual design (see Section 5.6.1.2).
Activities: Think about your system and UX research data and envision a conceptual design, including any metaphors, for how your overall system works. Try to communicate the designer's mental model, or a design vision, of how the system works.
Deliverables: Brief written descriptions of your conceptual design and/or a few presentation slides of the same to share with others.
Schedule: You decide how much time you can afford to give this.

14.4 SEGUE

In this chapter, we discussed the role of mental models and conceptual designs. In the next three chapters, we address design goals related to the pyramid of human needs.